

Conic Exercises: Lines, Ellipses, Parabolas, and Tangents

N-20220716

§1.1 Line and ellipse

The note continues the conic-section exercises. A line and an ellipse can have three possible positional relationships:

$$\begin{aligned}\Delta > 0 &: \text{two intersections,} \\ \Delta = 0 &: \text{tangent,} \\ \Delta < 0 &: \text{no real intersection.}\end{aligned}$$

This is obtained by substituting the line equation into the conic equation and examining the discriminant of the resulting quadratic.

For example, consider

$$y = 2x - 2, \quad \frac{x^2}{5} + \frac{y^2}{4} = 1.$$

Substitution gives

$$\frac{x^2}{5} + \frac{(2x - 2)^2}{4} = 1,$$

so

$$\frac{6}{5}x^2 - 2x = 0.$$

Thus

$$x = 0, \quad x = \frac{5}{3},$$

and the intersection points are

$$(0, -2), \quad \left(\frac{5}{3}, \frac{4}{3}\right).$$

§1.2 Tangents from a point to an ellipse

Let

$$C : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

and let $P = (0, 2)$. A line through P has equation

$$y = kx + 2.$$

Substituting into the ellipse gives a quadratic in x . Tangency occurs exactly when the discriminant is zero.

For the example on the page, this produces

$$k = \pm \frac{\sqrt{6}}{3},$$

so the two tangents are

$$y = \frac{\sqrt{6}}{3}x + 2, \quad y = -\frac{\sqrt{6}}{3}x + 2.$$

§1.3 Line and hyperbola

For the hyperbola

$$\frac{x^2}{2} - \frac{y^2}{2} = 1,$$

and the line

$$y = x - 1,$$

substitution gives

$$x^2 - (x - 1)^2 = 2.$$

Hence

$$2x - 1 = 2, \quad x = \frac{3}{2}, \quad y = \frac{1}{2}.$$

The line is not tangent; it meets the hyperbola at one point because the equation degenerates to a linear equation after substitution. This is a useful warning: for hyperbolas, “one intersection” need not mean tangency, because the line may be parallel to an asymptotic direction.

§1.4 Parabola tangent through a point

For the parabola

$$x^2 = \frac{1}{2}y,$$

we have

$$y = 2x^2.$$

A line through $P = (2, 6)$ may be written

$$y = kx + 6 - 2k.$$

Tangency means that

$$2x^2 = kx + 6 - 2k$$

has a repeated root. Thus

$$2x^2 - kx - 6 + 2k = 0,$$

and the discriminant condition is

$$k^2 - 4 \cdot 2(-6 + 2k) = 0.$$

Solving gives

$$k = 12, \quad k = -4.$$

The tangent lines are therefore

$$y = 12x - 18, \quad y = -4x + 14.$$

§1.5 A parabola and a circle

The second page considers the parabola

$$C : x^2 = 2py, \quad p > 0,$$

with focus F , and a circle

$$M : x^2 + (y + 4)^2 = 1.$$

The problem asks for p when the shortest distance between F and the circle is 4, and then asks for a maximal triangle area using two tangents from a point P on the circle.

Since the focus of $x^2 = 2py$ is

$$F = \left(0, \frac{p}{2}\right),$$

and the circle has center

$$O = (0, -4)$$

and radius 1, the distance from F to the circle is

$$FO - 1 = \left(\frac{p}{2} + 4\right) - 1 = \frac{p}{2} + 3.$$

Setting this equal to 4 gives

$$p = 2.$$

Hence the parabola is

$$x^2 = 4y.$$

For a tangent point $A = (x_1, x_1^2/4)$ on $x^2 = 4y$, the tangent has equation

$$xx_1 = 2(y + x_1^2/4),$$

or

$$y = \frac{x_1}{2}x - \frac{x_1^2}{4}.$$

Similarly for a second tangent point $B = (x_2, x_2^2/4)$. The two tangents meet at

$$P = \left(\frac{x_1 + x_2}{2}, \frac{x_1 x_2}{4}\right).$$

Using the condition that P lies on the circle gives an optimization problem for the area of triangle PAB . The computation in the page reduces it to a one-variable expression and obtains the maximum

$$[PAB]_{\max} = 20\sqrt{5}.$$

§1.6 An ellipse with prescribed quadrilateral area

The final exercise considers an ellipse

$$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b > 0,$$

passing through $A = (0, -2)$. Four vertices form a quadrilateral of area $4\sqrt{5}$. Since $A = (0, -2)$ lies on the ellipse, $b = 2$. The area condition gives

$$2ab = 4\sqrt{5},$$

so

$$a = \sqrt{5}.$$

Therefore

$$E : \frac{x^2}{5} + \frac{y^2}{4} = 1.$$

For the line through $P = (0, -3)$

$$\ell : y = kx - 3,$$

intersecting the ellipse at B, C , the equation is

$$\frac{x^2}{5} + \frac{(kx - 3)^2}{4} = 1.$$

This gives a quadratic in x . The condition that the line meet the ellipse in two points is

$$k > 1 \quad \text{or} \quad k < -1.$$

The problem then studies where the chords meet the line $y = -3$ and imposes

$$|PM| + |PN| \leq 15.$$

The computation on the page reduces this to

$$|k| \leq 3.$$

Combining conditions gives

$$k \in [-3, -1) \cup (1, 3].$$

§1.7 Discriminants distinguish tangency from asymptotic intersection

The page is a concentrated exercise in the algebraic method for conics: substitute a line into a conic, read the discriminant, and interpret the result geometrically. The only subtlety is that different conics behave differently: for ellipses, one intersection usually signals tangency; for hyperbolas, a line parallel to an asymptotic direction can produce a single finite intersection without being tangent.

§1.8 Tangency via discriminants

A line intersecting a conic usually produces a quadratic equation in the line parameter. Tangency occurs when the quadratic has a repeated root:

$$\Delta = 0.$$

This converts a geometric contact condition into algebra. It is the same multiplicity idea that appears in polynomial factorization.

§1.9 Polarity with respect to a conic

For a nondegenerate conic, a point outside the conic determines a polar line through its contact points with tangents. In matrix form, if the conic is

$$X^T A X = 0,$$

then the polar of P is

$$P^T A X = 0.$$

This gives a projective explanation of many tangent-chord relations.

§1.10 Area constraints and affine invariance

Some ellipse problems impose an area or quadrilateral condition. Area changes by the determinant of an affine map. Therefore one can often transform an ellipse to a circle, solve the symmetric problem, then transform back with the determinant factor.

§1.11 Tangency, polarity, and affine normalization

Line-conic exercises are controlled by three ideas: intersection multiplicity, polar duality, and affine normalization. The computations are coordinate expressions of these invariant geometric mechanisms.